Scalar multiplication

$v = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$

$c \in \mathbb{C} \quad cv = \begin{pmatrix} ca_1 \\ \vdots \\ ca_n \end{pmatrix}$

Dirac Notation (1930)

Op. vectors is equivalent matrix $v = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$

$|\psi\rangle = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$ ket (column vector)

$\langle\psi| = (a_1 \cdots a_n)$ bra (row vector)

$V, \mathbb{C}^n$ contains a special vector 0

$|0\rangle \in V \quad \text{or} \in \mathbb{C}^n$

$|0\rangle + |\omega\rangle = 0$

Basis (spanning set)

$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \cdots a_n \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$

$|v\rangle = \sum_i a_i |e_i\rangle$ where $|e_i\rangle$ are the basis

$V, \mathbb{C}^n \Rightarrow$ orthonormal is the basis

$V, \mathbb{C}^n$

$|\psi\rangle = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \quad |\varphi\rangle = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$

$|\psi\rangle + |\varphi\rangle = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix}$

Two linearly independent kets have the same exactly of vectors

$|\psi\rangle = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \quad |\varphi\rangle = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \quad \begin{vmatrix} a_1 & b_1 \\ \vdots & \vdots \\ a_n & b_n \end{vmatrix} = \begin{vmatrix} a_1 & b_1 \\ \vdots & \vdots \\ a_n & b_n \end{vmatrix}$

A set of vectors is linearly dependent

$\exists\ a_1 \ldots a_n \in \mathbb{C}$ st. $A \neq 0$

$a_1|v_1\rangle + a_2|v_2\rangle + \cdots a_n|v_n\rangle = 0$

$|v_1\rangle = \sum_{i=2}^{n} \frac{a_i}{a_1} |v_i\rangle$

$a_1|v_1\rangle + a_2|v_2\rangle + \cdots a_n|v_n\rangle = 0$

$V, \mathbb{C}^n \quad \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \quad V, \mathbb{C}^n \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Inner product

$|\psi\rangle \quad |\omega\rangle \in V$ build a scalar which is complex

$\langle\psi|\omega\rangle = \langle\omega|\psi\rangle = \sum_i a^* b_i$

$V, \mathbb{C}^n \quad |\psi\rangle = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \quad |\omega\rangle = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$

$\langle\psi|\omega\rangle = (a_1^* \cdots a_n^*) \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$

$\langle\psi|\omega\rangle = \sum_i a_i^* b_i$

$\langle\psi|\psi\rangle = \sum_i |a_i|^2 \geq 0$

Write some equipped with inner product is called an inner product space

$\langle\psi|\omega\rangle = \langle\omega|\psi\rangle^*$

$\langle\psi|\psi\rangle \geq 0$

$\langle\psi|\psi\rangle = 0 \Rightarrow$ these vectors are orthogonal to each other

Orthonormal vector

Normal vector $|\psi\rangle \Rightarrow \frac{|\psi\rangle}{\sqrt{\langle\psi|\psi\rangle}}$

$\langle\psi|\psi\rangle = 1$

$|\psi\rangle \Rightarrow$ normalized vector

$|\psi\rangle = \frac{|\psi\rangle}{\|\psi\|} \Rightarrow \|\psi\rangle = 1$

$|\psi\rangle, |\varphi\rangle$ such that $\langle\psi|\varphi\rangle = 0$

orthogonal to each other or orthonormal

$|\psi\rangle = \sum_i a_i |\varphi_i\rangle \quad |\varphi\rangle$ are basis vectors

$|\varphi\rangle = \sum_i w_i |\varphi_i\rangle \quad |\varphi\rangle$

$\langle\varphi|\psi\rangle = \sum_i \sum_j a_j^* w_i \langle\varphi_j|\varphi_i\rangle$

$\langle\varphi|\psi\rangle = \sum_i a_i^* w_i$

Orthonormal basis is a set of normal vectors

$\langle\varphi_i|\varphi_j\rangle = \delta_{ij}$

$\langle\varphi_i|\varphi_j\rangle = \delta_{ij}$

$V, \mathbb{C}^n$ orthonormal is each vectors $\Rightarrow$ orthonormal basis

$A = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |v\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \mathbb{C}^2 \quad |w\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \in \mathbb{C}^2$

$\langle v|w\rangle > 0$

$\langle v|v\rangle = 1$

$\langle v|w\rangle > 0$

$\langle v|v\rangle = 1$

$A|v\rangle = |v\rangle$

$A|v\rangle = |v\rangle$

$A_1 = \langle\varphi|A|\varphi\rangle \quad A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$

$A_2 = \langle\varphi|A|\psi\rangle$

$A_3 = \langle\psi|A|\varphi\rangle$

$A_4 = \langle\psi|A|\psi\rangle$

$A_1 = \langle\varphi|A|\varphi\rangle$

$A_2 = \langle\varphi|A|\psi\rangle$

$A = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |v\rangle\langle v| + |v\rangle\langle v|$

Outer product

$|v\rangle, |w\rangle \in V, W \qquad A : V \to W$

$|v\rangle\langle w| : V \to W \qquad A =$

$|v\rangle\langle w|(|v\rangle) = \langle w|v\rangle |v\rangle$

$= \lambda |w\rangle$

$A = \sum_i a_i |w_i\rangle\langle v| \quad |w_i\rangle \in W, |v\rangle \in V$

$|\psi\rangle = \sum_i \langle\varphi_i|\psi\rangle |\varphi_i\rangle$

$= \sum_i |\varphi_i\rangle\langle\varphi_i|\psi\rangle$

$= |\psi\rangle$

$\sum_i |\varphi_i\rangle\langle\varphi_i| = I$

$A = |v\rangle\langle v| = |v\rangle\langle v| \quad |v\rangle\langle v| = 0$

$A|v\rangle = (|v\rangle\langle v|)|v\rangle = |v\rangle\langle v|v\rangle$

$= |v\rangle\langle v|v\rangle$

$A|v\rangle = |v\rangle\langle v| + |v\rangle\langle v|)|v\rangle$

$= |v\rangle\langle v|$

$|v\rangle, |w\rangle$ orthonormal basis for V

$|v\rangle, |w\rangle$ orthonormal basis for W

$I_W = \sum_i |w_i\rangle\langle w_i| \qquad A$

$I_V = \sum_i |v_i\rangle\langle v_i| \qquad A\Sigma = A$

$A : V \to W$

$A = I_W A I_V$

$A = \sum_i \sum_j |w_i\rangle\langle w_i| A |v_j\rangle\langle v_j|$

$= \sum_{i,j} \langle w_i|A|v_j\rangle \quad |w_i\rangle\langle v_j|$

$= \sum_{i,j} A_{ij} |w_i\rangle\langle v_j|$

$W = V$

$A = \sum_{i,j} A_{ij} |v_i\rangle\langle v_j|$

Eigenvalues and Eigenvectors

$A : V \to V$

$A|v\rangle = \lambda|v\rangle \qquad \lambda \in \mathbb{C}$

equation $|v\rangle$ is eigenvector

equation λ is eigenvalue

A : find the eigenvalues using the characteristic polynomial

$C(\lambda) = \det(A - \lambda I)$

$A|v\rangle = \lambda|v\rangle$

$(A - \lambda I)|v\rangle = 0$

$\Rightarrow \det(A - \lambda I) = 0$

$C(\lambda) = \det(A - \lambda I)$

substitute to main polynomial to find the eigenvalues

A diagonal representation

$A = \sum_i \lambda_i |v_i\rangle\langle v_i| = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$

$|v_i\rangle$ form set of orthonormal vectors

$\langle v_i|v_j\rangle = \delta_{ij}$

An operator is diagonalizable if it has a diagonal representation

$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = |0\rangle\langle 0| - |1\rangle\langle 1|$

Ex. Pauli matrices

$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Find diagonal representation for X, Y, Z

$X = \sum_i \lambda_i |v_i\rangle\langle v_i| \qquad |+\rangle\langle +| - |-\rangle\langle -|$

Adjoints and Hermitian Operators

$A : V \to V \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad a, b, c, d \in \mathbb{C}$

$A^T = (A^*)^T \qquad A^* = \begin{pmatrix} a^* & b^* \\ c^* & d^* \end{pmatrix}$

$A^\dagger = (A^*)^T = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$

An operator is self-adjoint Hermitian if

$\langle v|A|w\rangle = \langle w|A^\dagger|v\rangle$

$A^\dagger$ is Hermitian conjugate of A

$(|v\rangle)^\dagger = \langle v|$

$\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}^\dagger = (v_1^* \cdots v_n^*)$

$(A^\dagger)^\dagger = A$

An operator is Hermitian if $A^\dagger = A$

$A^\dagger = A$

V^k consider k dimensional subspace $W \subseteq V$

$|1\rangle \ldots |k\rangle$ orthonormal vectors

$P = \sum_i^k |i\rangle\langle i|$ Projector onto subspace W

$|v\rangle \in V^k$

$P|v\rangle \in W$

$(|v\rangle)^\dagger = \langle v|$

$(|v\rangle\langle v|)^\dagger = |v\rangle\langle v|$

$\Rightarrow |v\rangle\langle v|$ is hermitian

$P = \sum_i |i\rangle\langle i| \Rightarrow P$ is hermitian

$P^\dagger = P$

$P^2 = P$ Exercise